Market Power and Firm Conduct

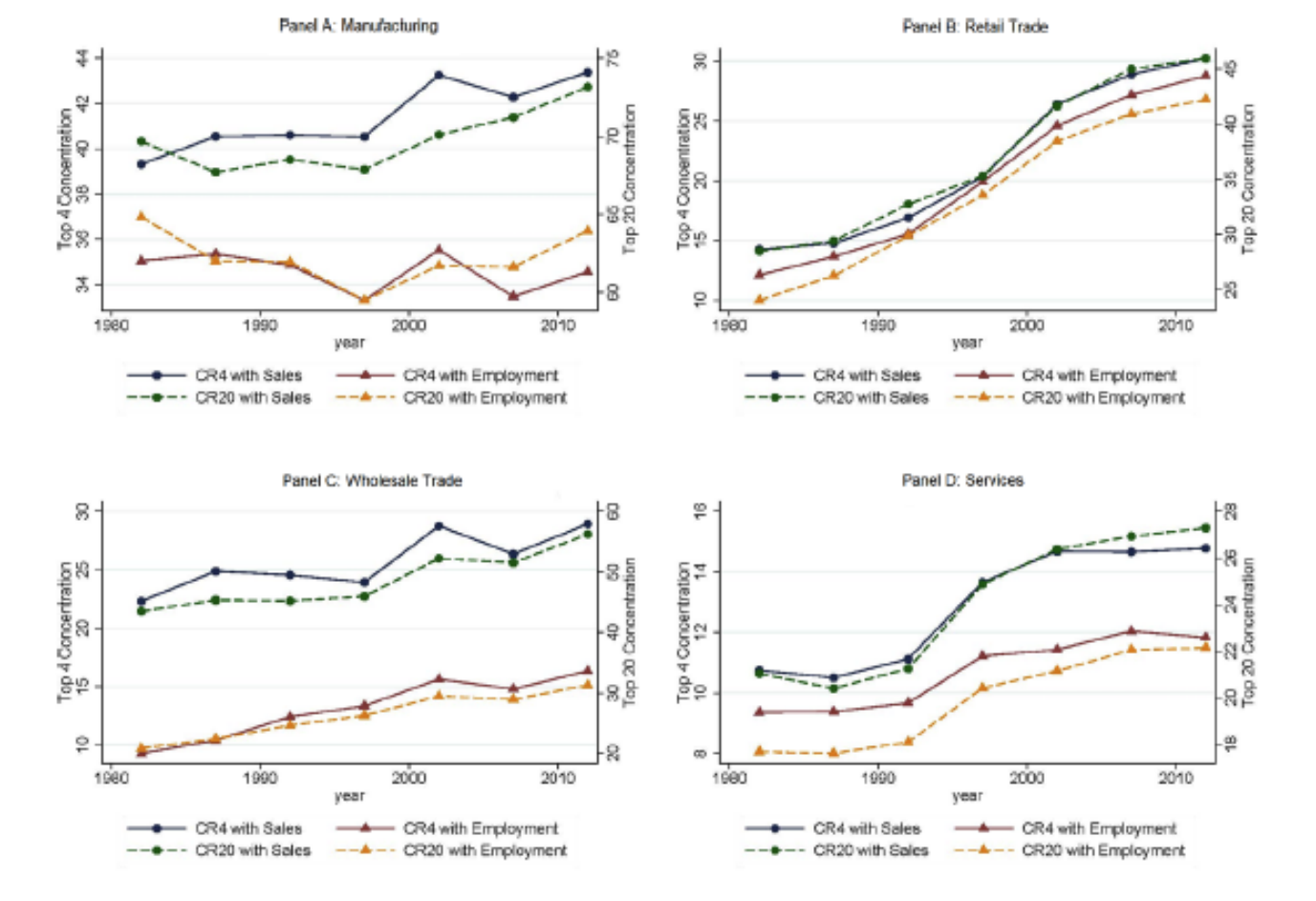
September, 2025

Introduction

- Market power has an increasing presence in the public policy debate. Examples include:
  - Rising corporate profit
  - Monopsony power in labor market
- A classical topic in the IO literature, somewhat "rediscovered" more recently.
- We will use this question as an organizing device to introduce the so-called New Empirical IO (NEIO) literature that started in the 1980s.
- More advanced/recent work will wait until we have covered demand and production estimation, stay tuned.

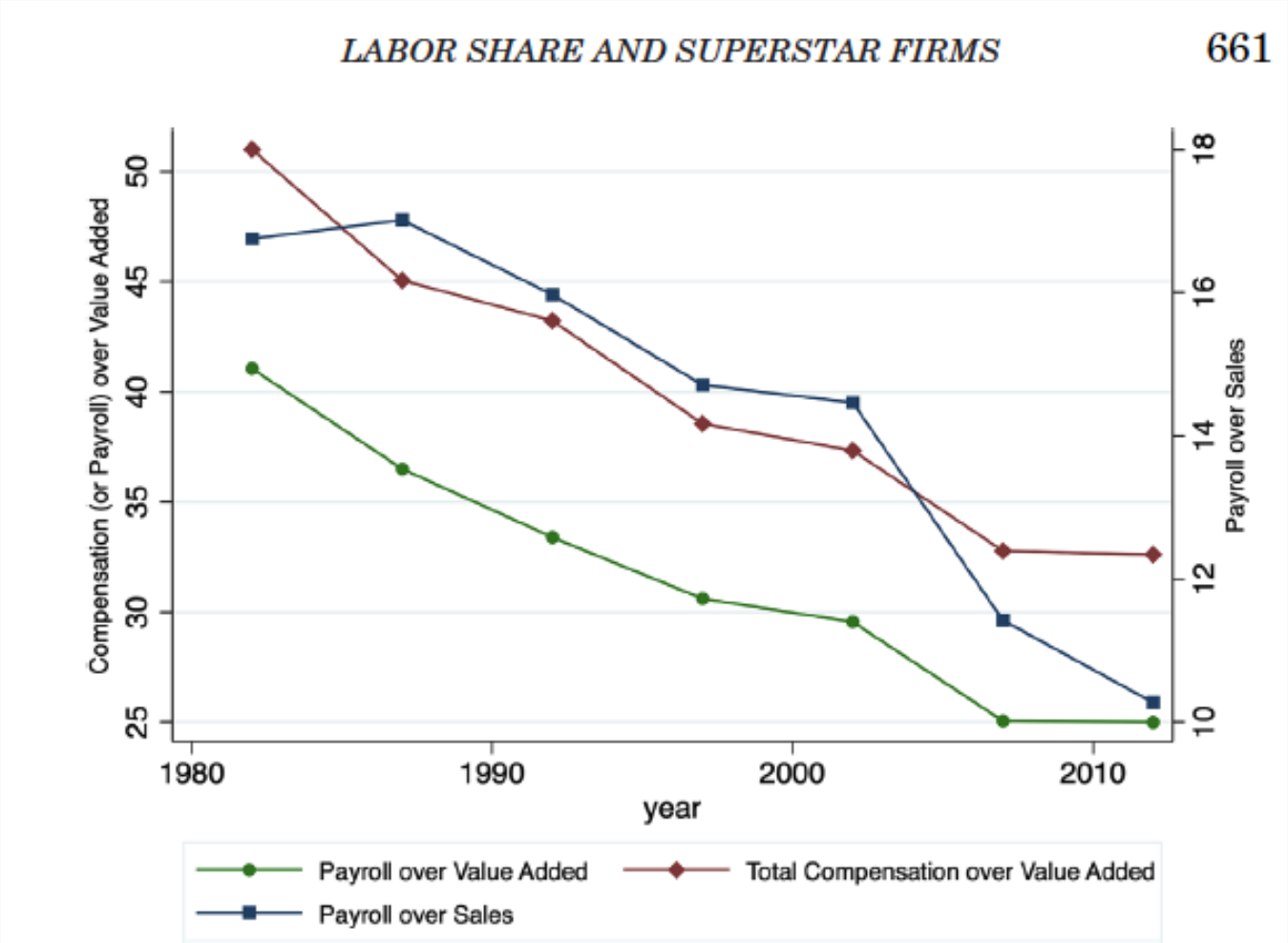
Some Motivating Facts

Rising conventional notion of industry concentration (Autor et al, QJE, 2020)



Some Motivating Facts

Declining labor share, in particular, in manufacturing (Autor et al, QJE, 2020)



Some Motivating Facts

"Forbidden regression" (Autor et al, QJE, 2020) shows negative corr. of the two overtime: market power?

TABLE III

INDUSTRY REGRESSIONS OF THE CHANGE IN THE PAYROLL-TO-SALES RATIO ON THE CHANGE IN CONCENTRATION, DIFFERENT SECTORS

	Stacked 5-year changes			Stacked 10-year changes		
	CR4 (1)	CR20 (2)	HHI (3)	CR4 (4)	CR20 (5)	HHI (6)
1 Manufacturing	−0.062***	−0.077***	−0.112***	−0.035	−0.034	−0.088**
$n = 2{,}328; 1{,}164$	(0.013)	(0.025)	(0.026)	(0.021)	(0.033)	(0.037)
2 Retail	−0.034*	−0.084**	−0.041	−0.043**	−0.067**	−0.068***
$n = 348; 174$	(0.020)	(0.037)	(0.025)	(0.018)	(0.029)	(0.023)
3 Wholesale	−0.038***	−0.040**	−0.084**	−0.037**	−0.036*	−0.064
$n = 336; 168$	(0.014)	(0.017)	(0.041)	(0.018)	(0.019)	(0.048)
4 Services	−0.091	−0.128***	−0.350***	−0.093	−0.137***	−0.377**
$n = 570; 258$	(0.057)	(0.039)	(0.084)	(0.070)	(0.042)	(0.156)
5 Utilities/Transport	−0.110***	−0.111**	−0.320***	−0.064	−0.096**	−0.226**
$n = 144; 48$	(0.031)	(0.050)	(0.082)	(0.044)	(0.038)	(0.098)
6 Finance	−0.221**	−0.252***	−0.567**	−0.236**	−0.274***	−0.723**
$n = 124; 62$	(0.084)	(0.091)	(0.208)	(0.095)	(0.084)	(0.295)
7 Combined	−0.077***	−0.088***	−0.150***	−0.060***	−0.076***	−0.118***
$n = 3{,}850; 1{,}901$	(0.017)	(0.022)	(0.028)	(0.018)	(0.023)	(0.032)

New Empirical Industrial Organization

Summarized in Bresnahan's Handbook of IO

- Firm's price-cost margin (performance) are not taken to be observables. I.e., the researcher needs to infer *marginal cost* from firm behavior.
- Individual industries are taken to have important idiosyncrasies. Institutional details $\rightarrow$ firm conduct and measurement strategy.
- Firm and industry conduct are viewed as *unknown* parameters to be estimated. A few examples, but we will see more:
  - Perfectly competitive
  - Cournot
  - Differentiated product Bertrand
  - Collusion
- Explicit Null hypothesis in testing conduct

Contrast with SCPP

This is in contrast to the SCPP where (1) use accounting data to directly measure performance – price-cost margin (2) summarize industry structure with a few observables (3) reduced-form relationship between structure and performance.

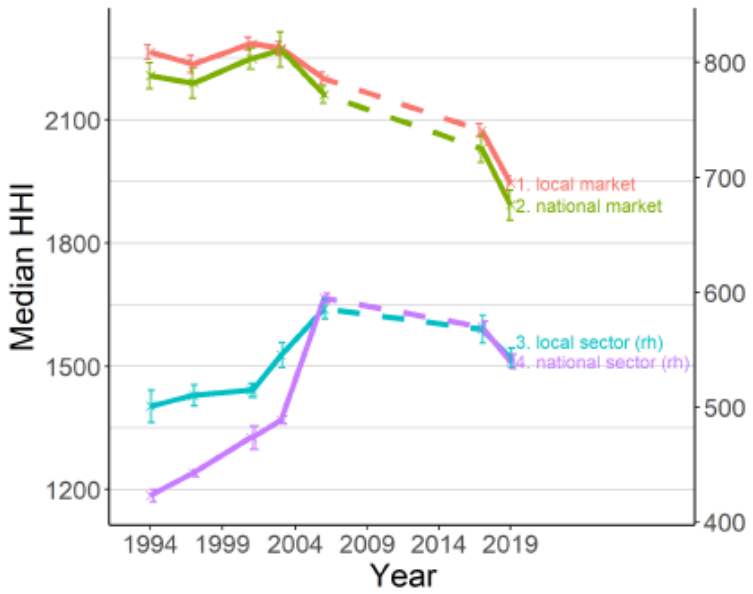
This reflects several deep doubts of IO economists

- whether accounting data can provide a reasonable measurement of *marginal cost* (or production technology).
- whether the relevant *market definition* overlaps with industrial classification of the firms.
- both the price-cost margin and the variables of "industry structure" are equilibrium outcomes – impacted by both demand and supply factors – hard to specify the valid IVs.

Market Definition Matters

Benkard et al (2021) uses bar code data and finds *decreasing* concentration in consumption goods market.

Figure 1: Median HHI over time, by market definition



Notes. Local markets are defined as product markets in each of 29 state groups. Sectors are defined by aggregating related national product markets. Product market measures are on the left hand side axis. Sector level measures are on the right hand side axis. Error bars are 95% CIs, based on standard errors from a nonparametric bootstrap.

Road-map from here

- Start with **homogeneous** product or **pure vertical differentiation** setting in the early 1980s.
  - Tools of (horizontally differentiated) demand and production estimations were not fully developed
  - Highlight the key identification intuitions
- Descriptive and quasi-random evidences: details matter.
  - Common Ownership
  - Hospital Mergers and Labor Market Power
- You should read a few more papers on differentiated products and input markets later once tooled up
  - Beer Joint Venture (Miller and Weinberg, 2017)
  - Chinese Tabacco Industry (Rubens, 2021)
  - Automobile revisited (Grieco et al, 2023)

Oligopoly Theory

Bresnahan starts with a general framework, with *homogeneous* product.

- Inverse demand:

$$P_t = D(Q_t, Y_t, \delta, \epsilon_{dt})$$

  where Y_t are demand shifters and ϵ_d is unobserved demand shock.

- Note that an auxiliary feature of NEIO is the application of econometric (identification) tools – "A typical NEIO paper is first and foremost an econometric model of an industry" (Bresnahan).

- Total cost:

$$C_{it} = C(Q_{it}, W_{it}, Z_{it}, \Gamma, \epsilon_{cit})$$

  where W_{it} are cost shifters, Z_{it} include SR fixed factors.
  $MC_{it} = C_1(Q_{it}, W_{it}, Z_{it}, \Gamma, \epsilon_{cit})$

- We now show firm's profit maximization problem (in this case, quantity setting) follows a *general relationship*

$$P_t = C_1(Q_{it}, W_{it}, Z_{it}, \Gamma, \epsilon_{cit}) - D_1(Q_t, Y_t, \delta, \epsilon_{dt})Q_{it}\theta_{it}$$

Deriving the Conduct Equation

Where does $P_t = C_1(\cdot) - D_1(\cdot) Q_{it} \theta_{it}$ come from? Firm i chooses Q_{it} to maximize profit, taking into account how its choice moves *industry* output $Q_t = \sum_j Q_{jt}$:

$$\max_{Q_{it}} \; \pi_i = P(Q_t)\, Q_{it} - C_i(Q_{it})$$

- FOC: $\dfrac{d\pi_i}{dQ_{it}} = P(Q_t) + Q_{it}\, P'(Q_t)\, \dfrac{dQ_t}{dQ_{it}} - C_i'(Q_{it}) = 0$
- Define the **conjectural variation** $\theta_{it} \equiv \dfrac{dQ_t}{dQ_{it}} = \dfrac{dQ_{it}}{dQ_{it}} + \sum_{j \neq i} \dfrac{dQ_{jt}}{dQ_{it}}$: how much firm i believes total output moves when it changes its own output (Bresnahan, 1989, eq. 13–14).
- Rearranging, with $P'(Q_t) \equiv D_1(\cdot)$:

$$P_t = C_i'(Q_{it}) - D_1(Q_t, Y_t, \delta, \epsilon_{dt})\, Q_{it}\, \theta_{it}$$

- $\theta_{it} = 0$: price-taking ($dQ_{jt}/dQ_{it} = -1$, rivals fully offset). $\theta_{it} = 1$: rivals hold quantity fixed ($dQ_{jt}/dQ_{it} = 0$) — the Cournot case below.

Examples

- The simplest case: perfect competitive $\theta_{it} = 0$.
- One-shot Nash Equilibrium with quantities as the strategic choice (Cournot): $\theta_{it} = 1$.
- Stackelberg leader: for the "leader firm" 1

$$P_t = C_1(Q_{1t}, W_{1t}, Z_{1t}, \Gamma, \epsilon_{c1t}) - D_1(Q_t, Y_t, \delta, \epsilon_{dt})Q_{1t}(1 + \theta_S)$$

θ_S is defined as the derivative of the sum of Cournot response function (of firm 2,...,I) with respect to Q_{1t}.

Examples

- Dominant firm model (Suslow, RAND): fringe firms are non-strategic and competitive, i.e.

$$P_t = C_1(Q_{ft}, W_{ft}, Z_{ft}, \Gamma, \epsilon_{cft}); Q_{ft} = \int_i Q_{ift} \equiv S(P_t, W_{ft}, Z_{ft}, \Gamma, \epsilon_{cft})$$

  In contrast, the strategic dominant firm take into account the impact of its output on P_t and subsequently supply schedule of fringe. Formally

$$\frac{\partial Q_f}{\partial Q_1} = S_1(\cdot)\frac{\partial P}{\partial Q_1} = S_1(\cdot)D_1(\cdot)(1 + \frac{\partial Q_f}{\partial Q_1})$$

  We have

$$\theta_S \equiv \frac{\partial Q_f}{\partial Q_1} = D_1(\cdot)S_1(\cdot)/(1 - D_1(\cdot)S_1(\cdot))$$

Econometric Identification

- One of the key feature of NEIO is its explicit argument of identification that is rooted in theory.
- Consider the mapping between data and parameters
  - Parameters of interest: demand δ, cost Γ, and conduct θ
  - Data: P_t, Q_{it} (or Q_t). Need to figure out what conditions on Y_t and W_{it} are needed to identify all parameters.
- Bresnahan (1989) summarizes various comparative statics (i.e. changes in economic environment) that can serve the purpose. But the most prevalent argument is based on demand Y_t
- Consider

$$P_t = D(Q_t, Y_t, \delta) + \epsilon_{dt} \quad \text{Demand}$$

$$P_t = C_1(Q_t, W_t, Z_t, \Gamma) - D_1(\cdot, Y_t, \delta)Q_t\theta \quad \text{Supply}$$

Econometric Identification

- Demand is identified as long as we have cost shifter W_t, so we can take δ^* as known. The tricky part is the supply

$$P_t = C_1(Q_t, W_t, Z_t, \Gamma) - D_1(\cdot, Y_t, \delta^*)Q_t\theta$$

- Two endogenous variables Q_t (in marginal cost) and $D_1(\cdot, Y_t, \delta^*)Q_t$.
- Cost shock ϵ_{cit} is generally correlated with quantity.

Alternative One: Rotation of Demand

- Absent Y_t in $D_1(\cdot)$, it is clear that the two endogenous variables are perfectly correlated – so we need more than just demand shifter.
- We must have Y_t contains IVs *separately* for both endogenous variables, in other words we need a demand IV that only affects $D_1(\cdot, Y_t, \delta^*)$ (slope of demand).
- This constitutes the famous argument of the "rotation of demand" (Bresnahan, 1982)
  - find a natural experiment that "rotates" the demand around the initial equilibrium (quantity E_1)
  - under perfect competition (and marginal cost curve MC^c) $\rightarrow$ demand/supply intersect at the same point
  - market power (and marginal cost curve MC^m) $\rightarrow$ marginal revenue $D_1(\cdot)Q$ shifted $\rightarrow$ new equilibrium point (quantity E_3)

Bresnahan (1982): Shifting Demand Alone Fails

Take linear demand and marginal cost (Bresnahan, 1982, eqs. 3–4):

$$Q = \alpha_0 + \alpha_1 P + \alpha_2 Y + \epsilon, \qquad MC = \beta_0 + \beta_1 Q + \beta_2 W$$

- Since demand is linear, $MR = P - Q/\alpha_1$. The general supply relation $P = \lambda(-Q/\alpha_1) + MC$ becomes

$$P = \beta_0 + \gamma Q + \beta_2 W + \eta, \qquad \gamma \equiv \beta_1 - \lambda/\alpha_1$$

- This equation is identified (only Q is included/endogenous, Y is excluded) — but γ bundles β_1 (cost curvature) and λ (market power) into one number.
- **Only shifting** demand (Y) up or down traces out the *same* γ whether $\lambda = 0$ (competition, MC^c steep) or $\lambda = 1$ (monopoly, MC^m flat) — a demand shift hits MC^c and MC^m at the same new point (P_2, Q_2).
- Intuition: without more structure, competition-with-steep-MC and monopoly-with-flat-MC are *observationally equivalent*.

Bresnahan (1982): Rotation Identifies Market Power

Add an interaction term so a variable Z can *rotate* demand around the equilibrium, not just shift it (eq. 7):

$$Q = \alpha_0 + \alpha_1 P + \alpha_2 Y + \alpha_3 PZ + \alpha_4 Z + \epsilon$$

- Demand slope is now $-1/(\alpha_1 + \alpha_3 Z)$: it depends on Z. The supply relation becomes (eq. 8)

$$P = \frac{-\lambda}{\alpha_1 + \alpha_3 Z} Q + \beta_0 + \beta_1 Q + \beta_2 W + \eta$$

- Writing $Q^* \equiv -Q/(\alpha_1 + \alpha_3 Z)$, there are now **two** included endogenous variables (Q, Q^*) and **two** excluded exogenous variables (Z, Y) $\Rightarrow \lambda$ is identified as the coefficient on Q^*.
- Graphically: rotating D around the initial point leaves a price-taker's equilibrium unchanged (still on MC^c), but shifts a monopolist's equilibrium (since MR has moved) — the two hypotheses become *observationally distinct*.

Bresnahan 1982

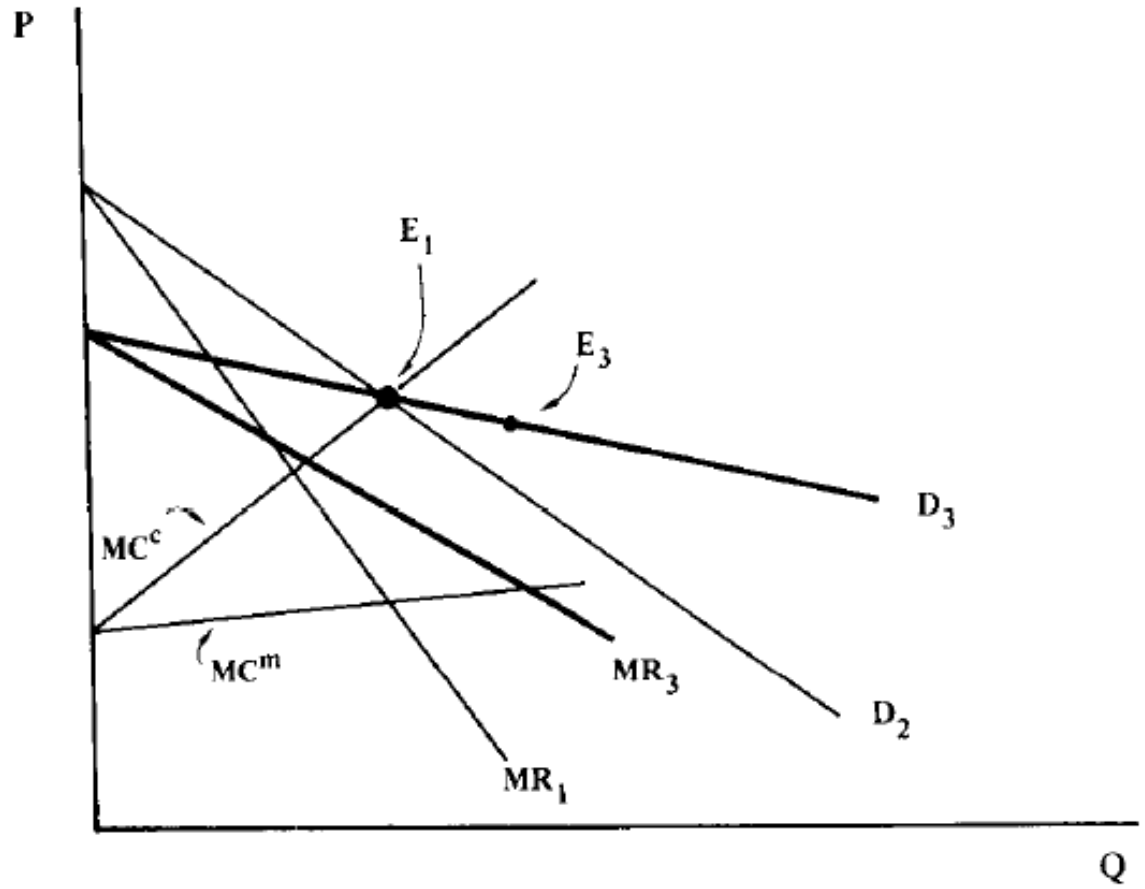


Fig. 2.

Other Alternatives

- Impose a functional form of total cost function (often flexible, say Translog).
  - Apply Shepard's Lemma such that input factor demand

$$X_t = C_W(Q_{it}, W_{it}, Z_{it}, \Gamma, \epsilon_{cit})$$

  - Directly plug in the implied $C_1(Q_{it}, W_{it}, Z_{it}, \Gamma, \epsilon_{cit})$ and attribute all the rest of variation to θ.
  - Challenge: MC needs to be estimated precisely – capital service demand is particularly challenging
- In the case of collusion (i.e. Porter 1983), the theory implies "supply shocks" – price wars and collusion specific θ_r and θ_c.
  - In separate "regimes", conditional on all exogenous observables (Z,W,Y), the different levels of P (and Q) identifies the *discrete* θ values.
  - If there were other "unobserved" shifters of supply, then it gets more tricky.

Porter (1983, RAND): Overview

- One of the classical papers that study railroad collusion
  - Joint Executive Committee: a legal cartel that coordinated market shares for eastward-bound grain shipments in the 1880s US
- Test the Green and Porter (1984) model: behavioral regime ("cooperative" vs. "non-cooperative/price war") varies over time
  - Imperfect information: firms only observe realized market price but not output choices of competitors
  - Firms cannot distinguish whether low market price is due to cheating or adverse demand shocks
  - Equilibrium trigger strategies involve "price war" episodes after prices fall below threshold (even caused by low demand, not cheating)
- Empirical problem: don't / imperfectly observe when "price war" occurs
  - To identify a true price war (a supply-side change) from price fluctuations, must rule out other demand and supply fluctuations

Porter (1983, RAND): Model

- N firms (railroads), homogeneous product (grain shipment)
- Market demand:

$$\log Q_t = \alpha_0 + \alpha_1 \log p_t + \alpha_2 L_t + U_{1t}, \quad Q_t = \sum_i q_{it}$$

  where L_t is an indicator if Great Lakes (main competition) were open.
- Firm i's cost function: $C_i(q_{it}) = a_i q_{it}^{\delta} + F_i$
- Firm i's pricing equation: $p_t \cdot (1 + \theta_{it}/\alpha_1) = MC_i(q_{it})$ and aggregating over firms, we have $p_t(1 + \theta_t/\alpha_1) = \sum s_i MC_i(q_{it})$ where $s_i = \frac{a_i^{1/(1-\delta)}}{\sum_j a_j^{1/(1-\delta)}}$.
- $\theta_t = 1$: collusion/monopoly, $\theta_t = \sum s_i^2$: Cournot, $\theta_t = 0$: Bertrand.
- Aggregate supply relation after some manipulation:

$$\log p_t = \log D - (\delta - 1) \log Q_t - \log(1 + \theta_t/\alpha_1)$$

  where $D = \delta(\sum_i a_i^{1/(1-\delta)})^{1-\delta}$, and the empirical version:

$$\log p_t = \beta_0 + \beta_1 \log Q_t + \beta_2 S_t + \beta_3 I_t + U_{2t}$$

  where I_t is an indicator for cooperative regime (i.e. $\theta \in \{\theta^c, \theta^p\}$)

Porter (1983): From Firm FOCs to Constant Market Shares

Where does $s_i = a_i^{1/(1-\delta)} / \sum_j a_j^{1/(1-\delta)}$ come from? In the Bertrand ($\theta_{it} = 0$) and collusive ($\theta_{it} = 1$) regimes, θ_{it} is the *same* for every firm, so each firm's FOC

$$p_t(1 + \theta_{it}/\alpha_1) = MC_i(q_{it}) = a_i \delta \, q_{it}^{\delta - 1}$$

has an identical left-hand side across firms i. Equating right-hand sides for any two firms i, j:

$$a_i \, \delta \, q_{it}^{\delta-1} = a_j \, \delta \, q_{jt}^{\delta-1} \quad \Longrightarrow \quad q_{it} = a_i^{1/(1-\delta)} \left[a_j^{-1/(1-\delta)} q_{jt} \right]$$

- So $q_{it} \propto a_i^{1/(1-\delta)}$ for every i, with the same proportionality constant across firms.
- Dividing by $Q_t = \sum_i q_{it}$, the constant cancels and market share is pinned down by relative cost shifters alone:

$$s_{it} = \frac{q_{it}}{Q_t} = \frac{a_i^{1/(1-\delta)}}{\sum_j a_j^{1/(1-\delta)}} \equiv s_i \quad \text{(constant over time)}$$

- Porter (1983) shows the same formula holds under Cournot conduct ($\theta_{it} = s_{it}$) as well — market shares are invariant to the collusive/non-cooperative regime, which is what lets a *single* aggregate supply relation nest all three cases.

Porter (1983, RAND): Model

- If I_t is known $\rightarrow$ estimate demand and supply equations using 2SLS
- If I_t is unknown, but assumed $I_t \overset{iid}{\sim} Bernoulli(\lambda)$, $(U_{1t}, U_{2t})' \overset{iid}{\sim} N(0, \Sigma)$, $\rightarrow$ estimate using maximum likelihood
  - Rewrite D&S: (note the exclusion restrictions, i.e. 0s)

$$By_t = \Gamma X_t + \Delta I_t + U_t,$$

where

$$y_t = \begin{pmatrix} \log Q_t \\ \log p_t \end{pmatrix}, \qquad X_t = \begin{pmatrix} 1 \\ L_t \\ S_t \end{pmatrix}, \qquad U_t = \begin{pmatrix} U_{1t} \\ U_{2t} \end{pmatrix}$$

and where

$$B = \begin{pmatrix} 1 & -\alpha_1 \\ -\beta_1 & 1 \end{pmatrix}, \qquad \Delta = \begin{pmatrix} 0 \\ \beta_3 \end{pmatrix}, \qquad \text{and} \qquad \Gamma = \begin{pmatrix} \alpha_0 & \alpha_2 & 0 \\ \beta_0 & 0 & \beta_2 \end{pmatrix}.$$

  - Likelihood function:

$$L = \prod_{t=1}^{T} f(y_t) = \prod_{t=1}^{T} \left[\lambda L(y_t | I_t = 1) + (1 - \lambda) L(Y_t | I_t = 0)\right]$$

where $L(y_t | I_t) = (2\pi)^{-1} |\Sigma|^{-\frac{1}{2}} ||B|| \cdot$

$$\exp\left\{ -\frac{1}{2}(By_t - \Gamma X_t - \Delta I_t)' \Sigma^{-1} (By_t - \Gamma X_t - \Delta I_t) \right\}$$

  - Algorithm: iterate between updating λ and collusive regime classifications $\{w_t\}$

Porter (1983, RAND): Data

TABLE 1 List of Variables*

GR	grain rate, in dollars per 100 lbs.
TQG	total quantity of grain shipped, in tons.
LAKES	dummy variable; =1 if Great Lakes were open to navigation; =0 otherwise.
PO	cheating dummy variable; =1 if colluding reported by *Railway Review;* =0 otherwise.
PN	estimated cheating dummy variable.
DM1	=1 from week 28 in 1880 to week 10 in 1883; =0 otherwise; reflecting entry by the Grand Trunk Railway.
DM2	=1 from week 11 to week 25 in 1883; =0 otherwise; reflecting an addition to New York Central.
DM3	=1 from week 26 in 1883 to week 11 in 1886; =0 otherwise; reflecting entry by the Chicago and Atlantic.
DM4	=1 from week 12 to week 16 in 1886; =0 otherwise; reflecting departure of the Chicago and Atlantic from the JEC.

* The sample is from week 1 in 1880 to week 16 in 1886.

- Weekly market-level price and quantity: GR and TQR
- Demand shifter (L_t in model): LAKES
- Supply shifters (I_t, S_t in model):
  - Collusion indicator: PO (reported), PN (estimated)
  - Structural dummies: DM1, DM2, DM3, DM4

Porter (1983, RAND): Results

TABLE 3 Estimation Results*

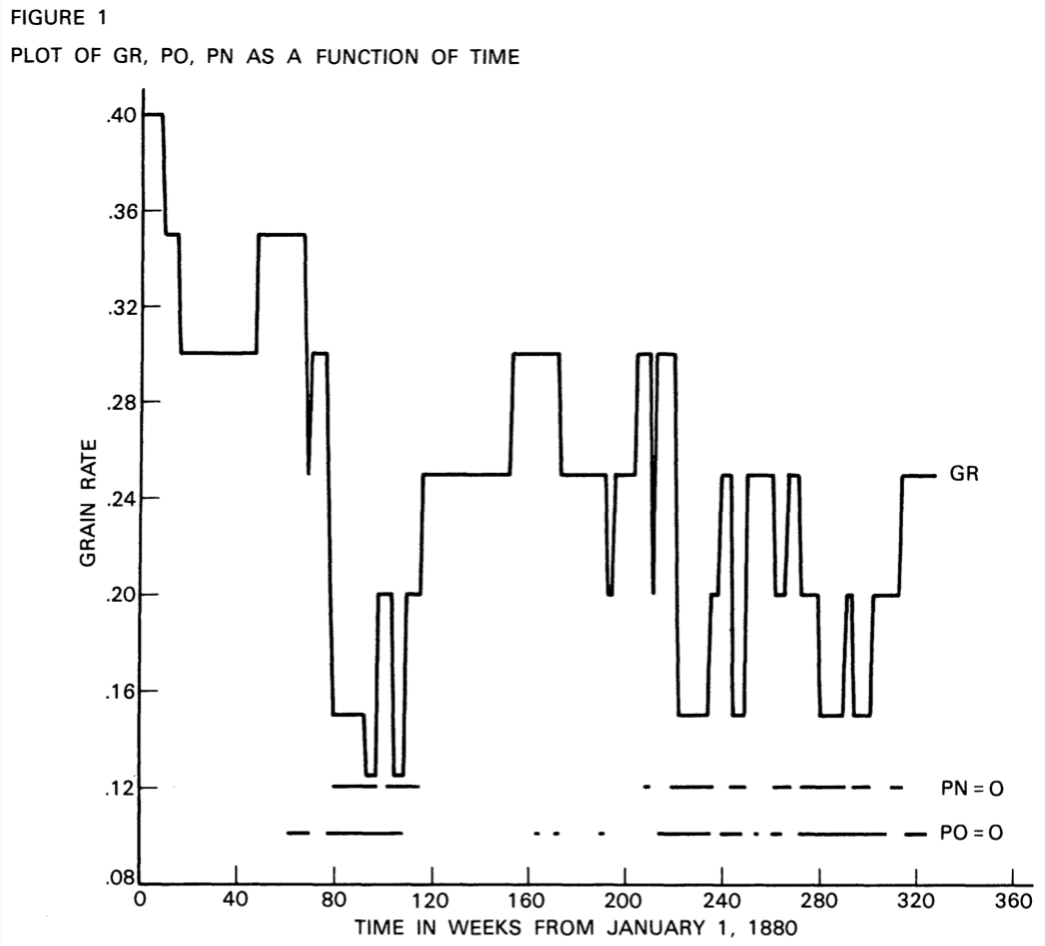
	Two Stage Least Squares (Employing *PO*)		Maximum Likelihood (Yielding *PN*)**	
Variable	Demand	Supply	Demand	Supply
C	9.169 (.184)	−3.944 (1.760)	9.090 (.149)	−2.416 (.710)
LAKES	−.437 (.120)		−.430 (.120)	
GR	−.742 (.121)		−.800 (.091)	
DM1		−.201 (.055)		−.165 (.024)
DM2		−.172 (.080)		−.209 (.036)
DM3		−.322 (.064)		−.284 (.027)
DM4		−.208 (.170)		−.298 (.073)
PO/PN		.382 (.059)		.545 (.032)
TQG		.251 (.171)		.090 (.068)
R^2	.312	.320	.307	.863
s	.398	.243	.399	.109

* Monthly dummy variables are employed. To economize on space, their estimated coefficients are not reported. Estimated standard errors are in parentheses.

** *PN* is the regime classification series $(\hat{I}, \ldots, \hat{I}_T)$. The coefficient attributed to *PN* is the estimate of β_3.

- Estimate two specifications
  - differ in whether reported collusion indicator is used
- LAKES: seasonal lake opening reduces demand
- DMs: structural dummies (entries) lowered market price
- PO/PN: prices are higher in "cooperative" regime
- Implied $\theta^c = 0.336 < 1$ in "cooperative" regime (assuming $\theta^p = 0$ in "non-cooperative" regime), consistent with Green and Porter (1984)

Porter (1983, RAND): Results



- Price wars occurred and were significant, but punishment periods vary in duration and magnitude

Bresnahan (1987): Empirical Puzzle

- **1955:** U.S. auto output $\uparrow \approx 45\%$ vs. 1954/56 while *quality-adjusted* prices $\downarrow$.
- Standard demand drivers (income, rates) too small; hedonic/quality indices still fall.
- **Hypothesis:** a *one-year* breakdown of tacit collusion $\Rightarrow$ **price war**.
- Strategy: estimate a differentiated-products model under two conduct regimes and discriminate with non-nested (Cox) tests.

Consumer Demand (Vertical Differentiation)

- Continuum of potential buyers indexed by taste v for quality x.
- Tastes: $v \sim \text{Uniform}(0, v_{\max})$ with density δ.
- If consumer buys model i with quality x_i and price p_i:

  $$U_i(v) = v\, x_i - p_i.$$

- If no new auto is bought, outside option (e.g., used car):

  $$U_0(v) = v\, y - \mathcal{E}.$$

- Consumer chooses the option minimizing $p - vx$ over all alternatives (including outside option).

Cutoffs and Choice Regions

- Order products by quality: $x_1 < x_2 < \cdots < x_n$.
- Indifferent consumer between adjacent h and i ($x_i > x_h$):

$$v_{h,i} = \frac{p_i - p_h}{x_i - x_h}.$$

- Outside vs. lowest-quality cutoff:

$$v_{0,1} = \frac{p_1 - \mathcal{E}}{x_1 - y}.$$

- Each car model serves an interval of v's bounded by neighboring cutoffs.

Demand System (Segments in v-space)

$$q_1 = \delta\,(v_{0,1} - v_{1,2}),$$
$$q_i = \delta\,(v_{i-1,i} - v_{i,i+1}), \quad i = 2, \ldots, n-1,$$
$$q_n = \delta\,(v_{\max} - v_{n-1,n}).$$

- $\frac{\partial q_i}{\partial p_i} < 0$. Cross-effects with neighbors reflect **closeness** in quality: the smaller $(x_i - x_{i-1})$, $(x_{i+1} - x_i)$, the stronger substitution.
- In the limit of equal qualities, close substitutes behave like (near) perfect substitutes.

Costs and Quality Proxy

- Per-model total cost: $C(x_i, q_i) = F_i + mc(x_i)\, q_i$ (fixed F_i sunk for pricing).
- Marginal cost convex in quality (exponential):

$$mc(x_i) = \exp\{\rho\, x_i\}, \qquad \rho > 0.$$

- Normalization for identification at bottom: impose $\mathcal{E} = mc(y)$.
- Quality mapping (latent x from observables z):

$$x_i = \beta_0 + \sum_k \beta_k \sqrt{z_{ik}}, z \in \{\text{length, weight, hp, cylinders, body-type}\}.$$

Profits and Conduct

- Per-model profits: $\pi_i = (p_i - mc(x_i))\, q_i(p)$.
- **Competition (N, Bertrand-Nash):** each firm sets prices for its own models taking rivals' prices as given.
- **Collusion (C):** set all prices to maximize joint profits $\sum_i \pi_i$.
- Cooperation structure via H:

$$H_{ij} = \begin{cases} 1, & \text{if } i \text{ and } j \text{ are jointly internalized,} \\ 0, & \text{otherwise.} \end{cases}$$

  Under N: H links same-firm neighbors; under C: H is all ones.

FOCs for Prices (Bertrand vs. Collusion)

Let $mc_i \equiv mc(x_i)$ and $\mathrm{MR}_i(p) = q_i(p) + (p_i - mc_i)\,\frac{\partial q_i}{\partial p_i}$.

- **Bertrand-Nash (no cross-internalization across owners):**

$$\frac{\partial \pi_i}{\partial p_i} = q_i + (p_i - mc_i)\,\frac{\partial q_i}{\partial p_i} = 0 \quad \Rightarrow \quad \mathrm{MR}_i = 0.$$

- **Collusion / internalized neighbors:**

$$\frac{\partial}{\partial p_i}\Big(\sum_j H_{ij}\,\pi_j\Big) = q_i + (p_i - mc_i)\,\frac{\partial q_i}{\partial p_i} + \sum_{j \neq i} H_{ij}\,(p_j - mc_j)\,\frac{\partial q_j}{\partial p_i} = 0.$$

- Under C, p_i internalizes effects on cooperating neighbors (especially strong for close substitutes).

Equilibrium and Solution

- Unknowns: $p_1, \ldots, p_n$ (then $q_1, \ldots, q_n$ implied).
- System: demand cutoffs + FOCs (Bertrand or Collusive).
- Because q_i are linear in prices, FOCs stack to a linear system in p.
- Solve simultaneously for $\mathrm{p}^*(\theta; z)$, $\mathrm{q}^*(\theta; z)$ given $\theta = (\beta, \rho, \delta, v_{\max}, y)$.

Estimation

- Observables: list prices (per model), production by model or aggregated cells, product characteristics z.
- **Aggregation:** if a reported cell j sums k_j physical models, assume

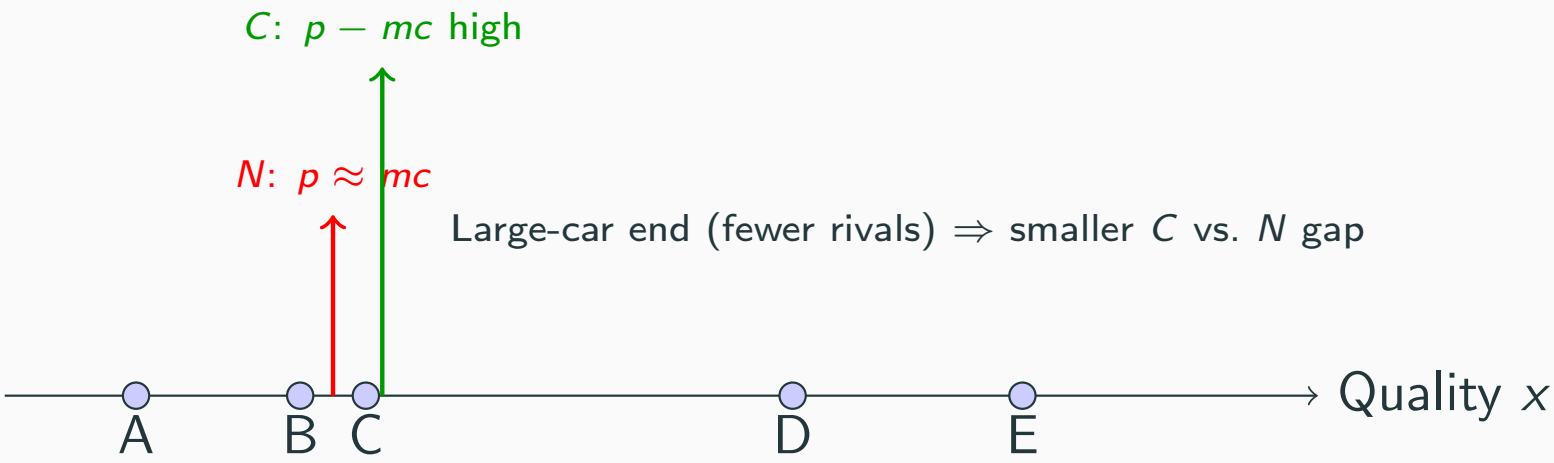
  $$\varepsilon_{q,j} \sim \mathcal{N}(0,\, k_j \sigma_q^2), \qquad \varepsilon_{p,i} \sim \mathcal{N}(0,\, \sigma_p^2),$$

  and independence across price/quantity parts.
- **Reduced-form errors:** additive to $\mathrm{p}^*, \mathrm{q}^*$.
- **Log-likelihood:** sum of normals with heteroskedastic quantity variances $k_j \sigma_q^2$.

Testing Conduct: Cox Non-Nested Tests

- Competing specifications:
  1. **Collusion** (joint-profit FOCs).
  2. **Nash-Competition** (Bertrand FOCs).
  3. "**Products**" model (each product its own firm).
  4. **Hedonic** (log-linear price & quantity eqs., firm dummies).
- Cox tests compare *pairs* of non-nested models; compute LR and its asymptotic variance under the maintained model.
- **Findings:** 1954 & 1956 $\Rightarrow$ Collusion not rejected; 1955 $\Rightarrow$ Collusion rejected, Nash not rejected.

Intuition by Segment: Close vs. Distant Substitutes



Quantitative Decomposition

- Fit 1955 parameters under N; then evaluate counterfactual 1955 prices/quantities under C.
- Result: only part of the 1955 surge attributable to demand; a large share comes from conduct change (price war).
- Structural supply (conduct) explains the anomaly.

Common Ownership: Backus et al 2021

- Share holder s has cash flow rights β_{fs} of the firm f they own.
- Investor is a "common owner" if $\beta_{fs} > 0$ for multiple f, the total profit for s is then $\nu_s = \sum_g \beta_{gs}\pi_g$.
- Each firm f puts Pareto weight γ_{fs} on its investor s, such that

$$Q_f(x_f, x_{-f}) = \sum_s \gamma_{fs} \sum_g \beta_{gs}\pi_g(x_f, x_{-f})$$

$$= \underbrace{\sum_s \gamma_{fs}\beta_{fs}\pi_f}_{\text{Own profit}} + \underbrace{\sum_s \gamma_{fs} \sum_{g\neq f} \beta_{gs}\pi_g}_{\text{Other profit}}$$

$$\propto \pi_f + \sum_{g\neq f} \underbrace{\left(\frac{\sum_s \gamma_{fs}\beta_{gs}}{\sum_s \gamma_{fs}\beta_{fs}}\right)}_{\kappa_{fg}(\gamma_f,\beta):\text{relative weights of firm } g \text{ to firm } f} \pi_g$$

- κ_{fg} is the counterpart of the conduct parameter in the 1980s NEIO literature.

Common Ownership: Decomposing κ_{fg}

While the true κ_{fg} are often hard to be identified directly from data (see the related work of the same authors on Cereal market.), literature often starts with $\gamma_{fs} = \beta_{fs}$ (i.e. one share one vote), $\kappa_{fg} = \langle \beta_f, \beta_g \rangle / \langle \beta_f, \beta_f \rangle$. Substitute the geometric definition of the inner product, $\langle x, y \rangle = \cos(x, y)\|x\|\|y\|$:

$$\kappa_{fg} = \frac{\cos(\beta_f, \beta_g)\, \|\beta_f\|\, \|\beta_g\|}{\|\beta_f\|^2} = \cos(\beta_f, \beta_g)\, \frac{\|\beta_g\|}{\|\beta_f\|}$$

- Define $\mathit{IHHI}_f \equiv \|\beta_f\|^2 = \sum_s \beta_{fs}^2$ (the Herfindahl index of firm f's *investors*). Then $\|\beta_g\|/\|\beta_f\| = \sqrt{\mathit{IHHI}_g / \mathit{IHHI}_f}$, giving

$$\kappa_{fg}(\beta) = \underbrace{\cos(\beta_f, \beta_g)}_{\text{overlapping ownership}} \times \underbrace{\sqrt{\mathit{IHHI}_g / \mathit{IHHI}_f}}_{\text{relative investor concentration}}$$

- Overlapping ownership alone drives up κ_{fg} only if investors are also relatively *more concentrated* in the rival g than in f — concentrated investors in f (large IHHI_f) make f place *less* weight on rivals' profits.
- Backus, Conlon and Sinkinson (2021) find relative investor concentration explains roughly 30–45% of the empirical variation in κ_{fg} (Table 3) — not just the "overlapping ownership" most of the literature emphasizes.

Common Ownership

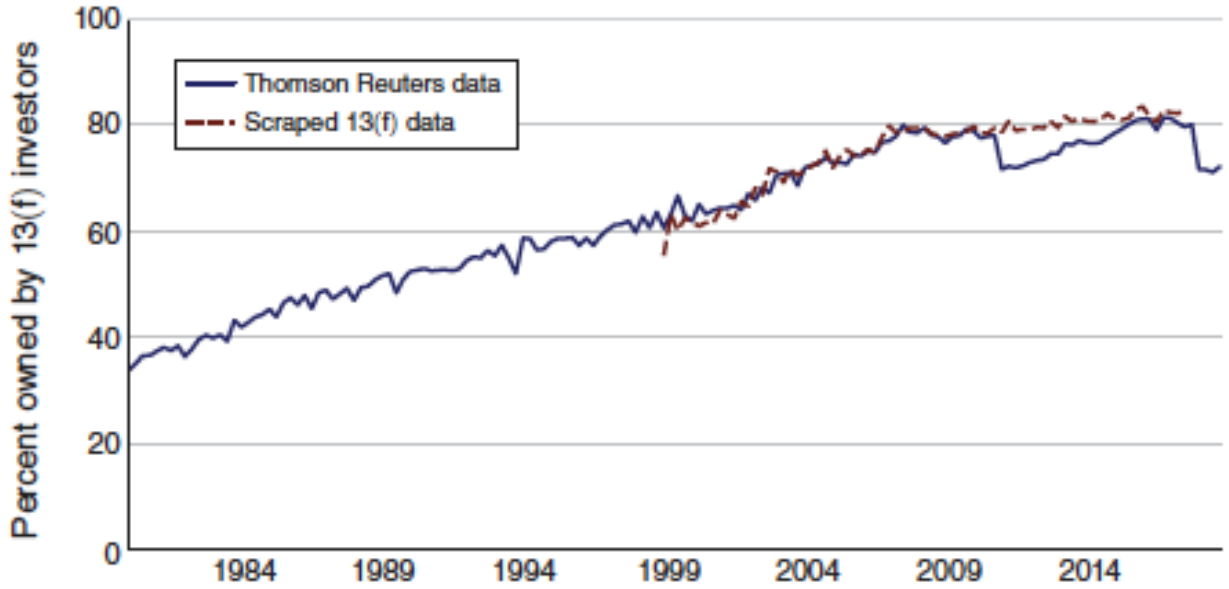


FIGURE 4. SHARE OF S&P 500 OWNED BY 13(f) MANAGERS

Notes: This figure depicts the average total share of a firm that is owned by managers filing 13(f). This corresponds to the institutional ownership share of the firm, and one hundred minus this number corresponds to what we are calling the retail share. We report the straight average across index constituents rather than a weighted average.

- Large rise of institutional investor ownership from 1980 to 2017

Common Ownership

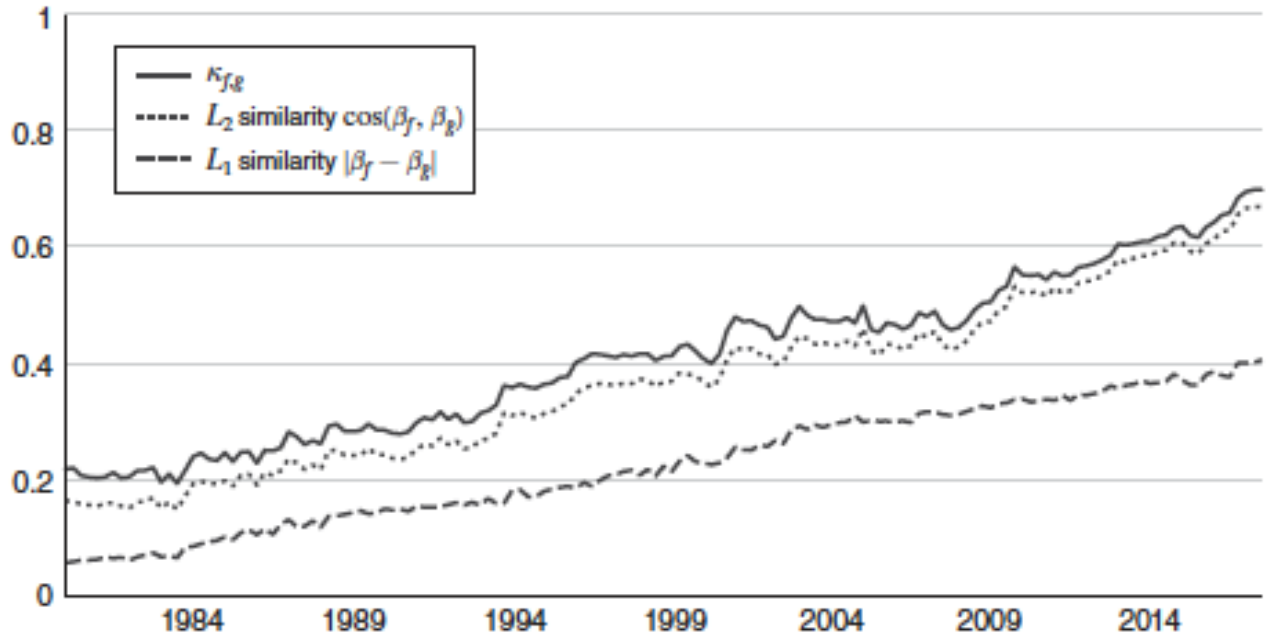


FIGURE 7. COSINE SIMILARITY AMONG INVESTORS

Notes: We report average similarity measures across all pairs of firms in each period. Note that similarity here is across firms (the vector β_f) rather than investors (the vector β_s), where the latter appears in Figure 8 below.

- Rising overlapping ownership due to indexing

Common Ownership

TABLE 3—DECOMPOSITION OF VARIANCE OF $\log \kappa$

	Overlapping ownership	Relative IHHI
Raw	68.67%	31.33%
Cross section	54.89%	45.11%
Time series	67.96%	32.04%
Panel	61.69%	38.31%

Notes: This table describes the attribution of variance according to the decomposition in equation (2). The raw sample is unmodified; the cross section is residualized on quarter fixed effects; the time series is residualized on ordered firm pair fixed effects; and, finally, the panel case is residualized on both quarter and pair fixed effects.

- Diversification of investor concentration also plays role

Merger and Wages: Prager and Schmitt, AER

TABLE 3—DIFFERENCE-IN-DIFFERENCES ESTIMATES

	Unskilled (1)	Skilled (2)	Nursing and pharmacy (3)
Post	0.005	−0.006	−0.007
	(0.005)	(0.008)	(0.006)
Observations	17,458	17,453	17,328
R^2	0.913	0.852	0.875

	Unskilled (4)	Skilled (5)	Nursing and pharmacy (6)
Post × first quartile ΔHHI	0.004	0.005	0.002
	(0.006)	(0.010)	(0.009)
Post × second quartile ΔHHI	0.007	−0.022	−0.001
	(0.009)	(0.016)	(0.010)
Post × third quartile ΔHHI	0.007	0.002	−0.019
	(0.008)	(0.021)	(0.014)
Post × fourth quartile ΔHHI	0.002	−0.041	−0.070
	(0.014)	(0.019)	(0.022)
Observations	17,458	17,453	17,328
R^2	0.913	0.853	0.875
H_0: no heterogeneity	0.978	0.105	0.016

Notes: All specifications include hospital and year fixed effects, plus the controls (log) one-bedroom rent, (log) population, (log) beds, (log) case mix index, percent Medicare, percent Medicaid, percent outpatient charges, (log) per capita income, percent unemployment, and percent of the population age 65 or older. Standard errors are clustered by hospital and observations are weighted by total inpatient discharges. The bottom row reports the *p*-value of a test of the null hypothesis that the post × ΔHHI quartile effects are equal to one another.

Merger and Wages

- Mergers slow down wage growth for health workers
- Key caveats
  - Worker skill is industry-specific
  - Rising in concentration is large
- Using out-of-market merger as placebo and finds no effect.
- Strong union mitigates such an effect.

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Further reading (differentiated products and input markets)

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